sac and is called as **endothelium** or **integumentary tapetum** (Example : Asteraceae). There are two types of ovule based on the position of the sporogenous cell. If the sporogenous cell is hypodermal with a single layer of nucellar tissue around it is called **tenuinucellate** type. Normally tenuinucellate ovules have very small nucellus. Ovules with subhypodermal sporogenous cell is called **crassinucellate** type. Normally these ovules have fairly large nucellus. Group of cells found at the base of the ovule between the chalaza and embryo sac is called **hypostase** and the thick -walled cells found above the micropylar end above the embryo sac is called **epistase**. The structure of ovule is given in Figure 1.7.

Types of Ovules

The ovules are classified into six main types based on the orientation, form and position of the micropyle with respect to funicle and chalaza. Most important ovule types are orthotropous, anatropous, hemianatropous and campylotropous. The types of ovule is given in Figure 1.8.

Orthotropous: In this type of ovule, the micropyle is at the distal end and the micropyle, the funicle and the chalaza lie in one straight vertical line. Examples: Piperaceae.

Anatropous: The body of the ovule becomes completely inverted so that the micropyle and funiculus come to lie very close to each other. This is the common type of ovules found in dicots and monocots.

Hemianatropous: In this, the body of the ovule is placed transversely and at right angles to the funicle. Example: Primulaceae.

Campylotropous: The body of the ovule at the micropylar end is curved and more or less bean shaped. The embryo sac is slightly curved. All the three, hilum, micropyle and chalaza are adjacent to one another, with the micropyle oriented towards the placenta. Example: Leguminosae

In addition to the above main types there are two more types of ovules they are,

Amphitropous: The distance between hilum and chalaza is less. The curvature of the ovule leads to horse-shoe shaped nucellus. Example: some Alismataceae.

Circinotropous: Funiculus is very long and surrounds the ovule. Example: Cactaceae

Megasporogenesis

The process of development of a megaspore from a megaspore mother cell is called **megasporogenesis**.

As the ovule develops, a single hypodermal cell in the nucellus becomes enlarged and functions as **archesporium**. In some plants, the archesporial cell may directly function as megaspore mother cell. In others, it may undergo a transverse division to form outer primary parietal cell and inner primary sporogenous cell. The parietal cell may remain undivided or divide by few periclinal and anticlinal divisions to embed the primary sporogenous cell deep into the nucellus. The primary sporogenous cell functions as a megaspore mother cell. The megaspore mother cell (MMC) undergoes meiotic division to form four haploid megaspores. Based on the number of megaspores that develop into the Embryo sac, we have three basic types of development: **monosporic, bisporic** and **tetrasporic**. The

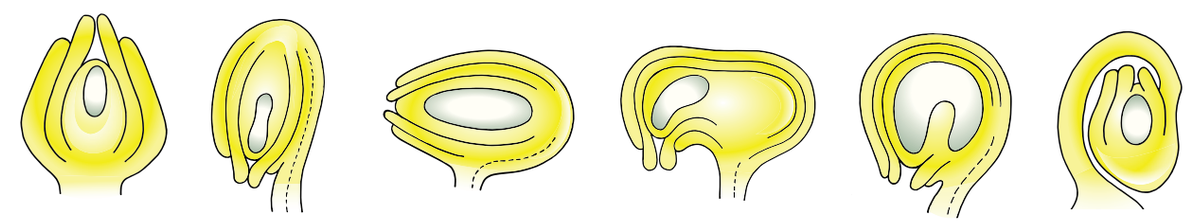


Figure 1.8 Types of ovule

TN_GOVT_BOTANY_XII_PAGES_001-032 CH 01.indd 11 23-12-2022 10:56:42